

CSC 392 ASSANGMENT PROJECT

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CAMPUS GUIDE APPLICATION

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1. Introduction

This section provides an overview of the entire Software Requirements Specification (SRS) for the Campus Guide Application (CGA). It defines the purpose, scope, definitions, references, and organization of this document, in accordance with IEEE Std 830-1998

1.1 Purpose

This SRS document specifies the software requirements for the Campus Guide Application (CGA) — a mobile application designed to help users navigate a university campus. The document is intended for:

- Software developers and designers responsible for building and maintaining the application
- Project managers and team leaders overseeing development and delivery
- Quality assurance engineers who will test and validate the system
- University stakeholders and administrators who will approve and deploy the system
- Future development teams who may extend or maintain the application

1.2 Scope

The software product to be developed is the Campus Student Guide App — a cross-platform mobile application for Android and iOS devices. The application will be named CampusGuide.

The CampusGuide application will:

- Provide students with an interactive, zoomable map of the entire campus
- Enable search and turn-by-turn walking navigation between any two campus locations
- Display Points of Interest (POI) such as the library, cafeteria, clinic, ATMs, and restrooms
- Provide real-time campus events, schedules, and announcements
- Offer accessibility routes for students with physical disabilities
- Function in offline mode using cached map data
- Display campus emergency contact numbers

The CampusGuide application will not:

- Replace the university's official student information or administration systems
- Collect, store, or transmit any Personally Identifiable Information (PII) of users
- Provide indoor room-level navigation (deferred to a future version)
- Support Augmented Reality (AR) navigation in the initial release

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification — this document
IEEE	Institute of Electrical and Electronics Engineers
API	Application Programming Interface
UI	User Interface
UX	User Experience
GPS	Global Positioning System
POI	Point of Interest (e.g. library, clinic, cafeteria)
MVP	Minimum Viable Product
AR	Augmented Reality
UAT	User Acceptance Testing
FR	Functional Requirement
NFR	Non-Functional Requirement
TBD	To Be Determined
HTTPS	HyperText Transfer Protocol Secure
TLS	Transport Layer Security
PII	Personally Identifiable Information
WCAG	Web Content Accessibility Guidelines
DFD	Data Flow Diagram
CGA	Campus Guide Application — the software product described in this SRS

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998..
- Google Maps Platform API Documentation. Google LLC. <https://developers.google.com/maps>
- Flutter SDK Documentation. Google LLC. <https://flutter.dev/docs>
- Firebase Documentation. Google LLC. <https://firebase.google.com/docs>
- WCAG 2.1 Guidelines. W3C. <https://www.w3.org/TR/WCAG21/>
- IEEE Std 610.12-1990. IEEE Standard Glossary of Software Engineering Terminology.
- Google Maps Platform Documentation. <https://developers.google.com/maps>
- Android Developer Documentation. <https://developer.android.com>
- Apple iOS Human Interface Guidelines. <https://developer.apple.com/design/human-interface-guidelines>

1.5 Overview

The remainder of this SRS is structured as follows:

- Section 2 — Overall Description: Background context including product perspective, functions, user characteristics, constraints, assumptions, and deferred requirements.
- Section 3 — Specific Requirements: Detailed functional and non-functional requirements organized by system feature, following the A.5 template of IEEE Std 830-1998, Annex A.
- Appendix A — Campus Location Data Structure: Defines the data model for Points of Interest.
- Appendix B — Requirements Traceability Matrix: Consolidated list of all requirements for verification purposes

2. Overall Description

This section provides a high-level description of the Campus Guide Application, its context within the university ecosystem, its key functions, intended users, and the constraints and assumptions that shape its requirements

2.1 Product Perspective

The CampusGuide application is a new, self-contained mobile application. It is not a replacement for any existing university system but complements them by providing a dedicated navigation layer for the physical campus. The application interfaces with the following external systems:

External System	Interface Type	Purpose
Google Maps API / Mapbox	Software Interface (REST API)	Render campus map tiles and compute walking routes
Firebase Realtime Database	Software Interface (WebSocket)	Store and retrieve live events and announcements
University Administration System	Data Interface (read-only)	Provide official building and room geodata
Device GPS	Hardware Interface	Determine the user's real-time location
Device Local Storage	Hardware Interface	Cache map data for offline mode
Android OS (API 28+)	Software Interface	Android platform support and permissions
iOS (13.0+)	Software Interface	iOS platform support and permissions

2.2 Product Functions

Function ID	Function Name	Brief Description
F-01	Interactive Campus Map	Zoomable, scrollable map of the full campus with POI markers
F-02	Location Search	Search bar with autocomplete to find buildings, rooms, and POIs
F-03	Turn-by-Turn Navigation	Walking directions from user location to any campus destination
F-04	Accessibility Routes	Alternate routes using ramps, elevators, and level pathways
F-05	Points of Interest	Display and detail view for library, cafeteria, clinic, ATMs, restrooms
F-06	Campus Events	Real-time list of upcoming campus events with details

Function ID	Function Name	Brief Description
F-07	Announcements	Urgent campus-wide notices displayed as home-screen banners
F-08	Offline Mode	Core map and POI functionality available without internet
F-09	Emergency Contacts	One-tap access to campus security, clinic, and emergency services

2.3 User Characteristics

User Group	Description	Technical Expertise	Primary Need
New / Freshman Students	First-semester students with no knowledge of campus layout	Low	Full navigation guidance and orientation
Transfer Students	Students joining from other institutions	Low to Medium	Quick campus layout understanding
Returning Students	General student population for daily use	Medium	Fast search and route to specific rooms
Visitors & Guests	Parents, event attendees, prospective students	Low	Simple wayfinding to key venues
Students with Disabilities	Students requiring accessible routes	Low to Medium	Accessible route guidance to all locations

2.4 Constraints

- **Device Compatibility:** The application must support Android 10+ and iOS 14+ to cover the majority of devices in use by the target user groups.
- **GPS Accuracy Limitation:** Standard mobile GPS has accuracy of 3–10 metres outdoors; indoor positioning accuracy will be lower and dependent on Wi-Fi availability.
- **Campus Map Currency:** The accuracy of navigation depends entirely on the quality and currency of the campus map data provided by the university. Outdated data is an operational, not software, risk.
- **Network Dependency:** Real-time features (live map updates, directory sync) require an internet connection. Core navigation must degrade gracefully when offline.
- **Data Privacy:** The application must not store or transmit a user's location history to any external server without explicit user consent, in compliance with applicable privacy regulations.
- **Accessibility:** The application must meet WCAG 2.1 Level AA standards to be usable by students and staff with disabilities.
- **Platform Guidelines:** The application must conform to Google Play Store and Apple App Store submission and design guidelines.

2.5 Assumptions and Dependencies

- University administration will provide official campus geodata before Week 6 of the project.
- Google Maps API free-tier quota (28,500 map loads/month) is sufficient for the pilot phase.
- Firebase platform availability is required for real-time events; unavailability triggers a graceful 'Service unavailable' notice.
- Device GPS is available; users will be prompted to enable location services if not.
- Target devices have a minimum 4.5-inch screen and at least 2 GB RAM.
- Flutter SDK (version 3.x) remains stable throughout the development period.

2.6 Apportioning of Requirements

Deferred Feature	Reason for Deferral	Target Version
AR Navigation	Requires additional hardware sensor integration beyond MVP scope	v2.0
Indoor Room-Level Navigation	Requires BLE beacon infrastructure not yet available	v2.0
Multi-Language Support	Translation resources require additional time and budget	v2.0
User Accounts & Saved Locations	Introduces PII concerns requiring additional privacy review	v2.0
Push Notifications for Events	Requires user consent and notification permission management	v2.0

3. Specific Requirements

This section specifies all detailed functional and non-functional requirements for the Campus Guide Application. Requirements are organized by system feature following the A.5 template of IEEE Std 830-1998, Annex A. Each requirement is uniquely identified, stated in verifiable terms, and shall be traceable to test cases during verification

3.1 External Interface Requirements

3.1.1 User Interfaces

[NFR-UI-01] The application shall provide a native mobile user interface that follows Android Material Design guidelines (Android) and Apple Human Interface Guidelines (iOS).

[NFR-UI-02] The main screen shall display the interactive campus map as the primary interface element, occupying at least 75% of the screen area.

[NFR-UI-03] The interface shall include a persistent search bar at the top of the map screen allowing users to search for any campus location at any time.

[NFR-UI-04] All text in the application shall be a minimum of 14sp font size to ensure readability on small mobile screens.

[NFR-UI-05] The application shall support both light and dark display modes, following the device's system-level display preference.

[NFR-UI-06] The application shall be fully operable using one hand on a standard smartphone (screen sizes 4.7 to 6.9 inches)

3.1.2 Hardware Interfaces

[NFR-HW-01] The application shall interface with the device's GPS/Location Services API to retrieve the user's current position for real-time map positioning and navigation.

[NFR-HW-02] The application shall interface with the device's camera hardware to enable QR code scanning for location lookup.

[NFR-HW-03] The application shall interface with the device's compass/magnetometer sensor to support map orientation relative to the user's heading.

3.1.3 Software Interfaces

Interface Name	Product	Version	Purpose
Map Service	Google Maps API / Mapbox SDK	Latest stable	Render map tiles and compute walking routes
Backend	Firebase Realtime Database	Latest stable	Store and retrieve events and announcements
Framework	Flutter SDK	3.x	Cross-platform mobile app development
Android Runtime	Android SDK	API 28+	Android OS-level permissions and APIs

Interface Name	Product	Version	Purpose
iOS Runtime	iOS SDK	13.0+	iOS OS-level permissions and APIs

3.1.4 Communications Interfaces

- **[CI-01] The system shall** communicate with the Google Maps API exclusively over HTTPS using REST protocol.
- **[CI-02] The system shall** communicate with Firebase over a secure WebSocket connection (wss://) for real-time updates.
- **[CI-03] The system shall** detect loss of network connectivity within 3 seconds and switch to offline mode automatically without crashing.
- **[CI-04] The system shall** retry failed network requests up to 3 times with exponential back-off before displaying an error.

3.2 Functional Requirements

Functional requirements are organised per the A.7 template: information flows (3.2.1), process descriptions (3.2.2), data construct specifications (3.2.3), and data dictionary (3.2.4).

3.2.1 Information Flows

3.2.1.1 Data Flow Diagram 1 — Campus Navigation Flow

Data Entity	From	To	Description
Search Query	User (text input)	Search Engine Process	Building name, room number, or POI keyword (2–100 chars)
Search Results	Search Engine Process	UI Results List	Up to 10 ranked matching campus locations
Selected Destination	User (tap on result)	Navigation Process	The target campus location chosen by the user
User Location	Device GPS	Navigation Process	Real-time GPS coordinates (latitude, longitude)
Map Tiles	Google Maps API	Map Display Process	Visual map segments for the campus area
Campus Geodata	University DB / Cache	Search + Navigation	Building names, coordinates, walkway paths
Route Data	Navigation Process	UI Map Overlay	Step-by-step walking instructions
ETA	Navigation Process	UI Overlay Panel	Estimated walking time in minutes

3.2.1.2 Data Flow Diagram 2 — Events and Announcements Flow

Data Entity	From	To	Description
Event Record	University Admin	Firebase Database	Event title, description, date/time, and location
Announcement Record	University Admin	Firebase Database	Urgent notice text, priority flag, and expiry date
Event Feed	Firebase Database	Events Display Process	Sorted list of upcoming events in real time
Urgent Banner	Announcements Process	Home Screen UI	High-priority notice displayed as a home-screen banner

3.2.1.3 Data Flow Diagram 3 — Offline Mode Flow

Data Entity	From	To	Description
Cached Map Tiles	Device Local Storage	Map Display Process	Previously downloaded campus map tile images
Cached POI Records	Device Local Storage	POI Display Process	Locally stored points of interest and coordinates
Connectivity Status	Device Network Monitor	Offline Mode Controller	Boolean flag indicating internet availability
Offline Mode Flag	Offline Mode Controller	All Network Processes	Disables Firebase; activates cached data sources

3.2.2 Process Descriptions

3.2.2.1 Process 1 — Campus Map Display

Input: User Location (GPS), Cached/Live Map Tiles, Campus Geodata, POI Records.

Algorithm: On launch, the system fetches GPS coordinates and requests map tiles from Google Maps API (or local cache if offline). The map renders with POI markers overlaid at their coordinates.

Affected Data Entities: Map Tiles (consumed), POI Records (read), Rendered Map (output to UI).

- **[P01] The system shall** load and display the interactive campus map within 3 seconds on a standard Wi-Fi connection.
- **[P02] The system shall** centre the map view on the user's current GPS location by default upon every launch.
- **[P03] The system shall** display POI markers for: library, cafeteria, health clinic, ATMs, restrooms, and parking areas.
- **[P04] The system shall** allow the user to tap any POI marker to view its name, category, and a brief description.

- **[P05] The system shall** support pinch-to-zoom and drag-to-pan gestures on the campus map.

3.2.2.2 Process 2 — Location Search

Input: Search Query (user text), Campus Geodata (local index).

Algorithm: As the user types, the system queries a locally stored campus location index for partial text matches. Results are ranked by relevance and proximity. Up to 10 results are returned.

Affected Data Entities: Search Query (consumed), Campus Geodata (read), Search Results (output).

- **[P06] The system shall** accept a text input search query and return matching results within 2 seconds of query submission.
- **[P07] The system shall** begin displaying autocomplete suggestions after the user types 2 or more characters.
- **[P08] The system shall** display a 'No results found' message when the query matches no known campus location.
- **[P09] The system shall** allow the user to select a result from the list to view its location on the map or start navigation.

3.2.2.3 Process 3 — Turn-by-Turn Navigation

Input: User Location (GPS), Selected Destination, Campus Geodata (walkway graph).

Algorithm: The system computes the shortest walkable route using the campus walkway network graph. The route is rendered as a highlighted polyline with step-by-step instructions.

Affected Data Entities: Route Data (generated), User Location (updated), Walkway Graph (read).

- **[P10] The system shall** calculate and display a walking route between any two campus locations within 2 seconds of destination selection.
- **[P11] The system shall** render the walking route as a highlighted polyline overlaid on the campus map.
- **[P12] The system shall** display step-by-step walking instructions in a scrollable list below the map.
- **[P13] The system shall** update the route in real time if the user deviates more than 20 metres from the suggested path.
- **[P14] The system shall** display the estimated walking time and total distance for the current route.
- **[P15] The system shall** provide an 'Accessible Route' toggle restricting navigation to ramps, elevators, and level pathways only.

3.2.2.4 Process 4 — Events and Announcements

Input: Event Feed (Firestore), Announcement Records (Firestore).

Algorithm: On opening the Events screen, the system retrieves the event list from Firestore, sorts by date ascending, and renders in a scrollable list. Urgent records appear as home-screen banners.

Affected Data Entities: Event Feed (consumed), Announcement Records (consumed), Events List UI (output).

- **[P16] The system shall** display a chronologically sorted list of upcoming campus events on the Events screen.
- **[P17] The system shall** display event details — title, date, time, location, and description — when a user taps on an event.
- **[P18] The system shall** display urgent announcements as a dismissible banner at the top of the home screen.
- **[P19] The system shall** refresh the events list automatically when the user navigates to the Events screen.

3.2.2.5 Process 5 — Offline Mode

Input: Connectivity Status (device), Cached Map Tiles, Cached POI Records.

Algorithm: The network monitor detects connectivity loss and raises an Offline Mode Flag. Processes switch to cached local sources. Live-data features are disabled with informative messages.

Affected Data Entities: Connectivity Status (read), Offline Mode Flag (raised/cleared), Cached Data (activated).

- **[P20] The system shall** cache the core campus map tiles and all POI records to device storage on first successful application launch.
- **[P21] The system shall** display a clearly visible 'Offline Mode' status indicator in the application header when no internet is detected.
- **[P22] The system shall** continue to display the cached campus map and POI markers while in offline mode.
- **[P23] The system shall** disable the Events and Announcements features during offline mode and display a 'Content unavailable offline' message.

3.2.2.6 Process 6 — Emergency Contacts

Input: Static Emergency Contact Records (hardcoded at build time, no network call required).

Algorithm: The Emergency screen displays a pre-loaded static contact list. The user taps a contact to trigger a native phone call.

Affected Data Entities: Emergency Contact Records (read), Phone Dialler (invoked).

- **[P24] The system shall** display a static list of emergency contact numbers: campus security, health clinic, fire services, and police.
- **[P25] The system shall** allow the user to initiate a phone call to any emergency contact with a single tap using the device's native dialler.
- **[P26] The system shall** make the Emergency Contacts screen accessible at all times, including during offline mode.

3.2.3 Data Construct Specifications

3.2.3.1 Construct 1 — CampusLocation

Record Type: Represents any named, navigable location on the university campus.

Field Name	Data Type	Description
locationID	String (UUID)	Unique identifier for the campus location

Field Name	Data Type	Description
locationName	String (max 100 chars)	Human-readable name (e.g. 'Science Block B, Room 104')
category	Enum	BUILDING, CLASSROOM, LAB, LIBRARY, CAFETERIA, CLINIC, ATM, RESTROOM, PARKING, OTHER
latitude	Float (6 d.p.)	GPS latitude coordinate of the location centroid
longitude	Float (6 d.p.)	GPS longitude coordinate of the location centroid
floorLevel	Integer	Floor number: 0 = ground, negative = basement
isAccessible	Boolean	True if reachable via an accessible (ramp/lift) route
description	String (max 300 chars)	Brief description of the location and its services

3.2.3.2 Construct 2 — EventRecord

Record Type: Represents a scheduled campus event or urgent announcement stored in Firebase.

Field Name	Data Type	Description
eventID	String (UUID)	Unique identifier for the event
title	String (max 100 chars)	Short title of the event
description	String (max 500 chars)	Full description and any relevant instructions
eventDate	DateTime (ISO 8601)	Scheduled date and start time of the event
locationName	String (max 100 chars)	Campus location name where the event is held
isUrgent	Boolean	If true, displayed as a home screen announcement banner
expiryDate	DateTime (ISO 8601)	Date/time after which the record is removed from the feed

3.2.3.3 Construct 3 — RouteRecord

Record Type: Represents a computed walking route between two campus locations.

Field Name	Data Type	Description
routeID	String (UUID)	Unique identifier for the computed route instance
originLocation	CampusLocation	The starting point of the navigation route

Field Name	Data Type	Description
destinationLocation	CampusLocation	The end point of the navigation route
steps	Array<String>	Ordered list of natural language navigation instructions
polylineCoords	Array<LatLng>	Ordered GPS coordinates forming the route path on the map
distanceMeters	Float	Total route distance in metres
estimatedTimeMin	Integer	Estimated walking time in minutes (5 km/h average)
isAccessible	Boolean	True if the route is fully accessible (no stairs)

3.2.3.4 Construct 4 — EmergencyContact

Record Type: Represents a static emergency contact entry stored locally in the application.

Field Name	Data Type	Description
contactID	String	Unique identifier for the contact entry
name	String (max 80 chars)	Display name of the service (e.g. 'Campus Security')
phoneNumber	String (E.164 format)	Phone number in international format for direct dialling
category	Enum	SECURITY, HEALTH, FIRE, POLICE, OTHER
description	String (max 150 chars)	Brief description of the service and when to call it

3.2.4 Data Dictionary

Data Element	Representation	Units / Format	Precision / Accuracy	Range
latitude	Float	Decimal degrees	6 decimal places	-90.000000 to 90.000000
longitude	Float	Decimal degrees	6 decimal places	-180.000000 to 180.000000
estimatedTimeMin	Integer	Minutes	Nearest whole minute	1 to 60
distanceMeters	Float	Metres	1 decimal place	0.0 to 10,000.0
searchQuery	String	UTF-8 text	N/A	2 to 100 characters
mapZoomLevel	Integer	Google Maps zoom units	Integer	14 (overview) to 20 (room level)

Data Element	Representation	Units / Format	Precision / Accuracy	Range
eventDate	DateTime	ISO 8601	Millisecond precision	Future dates; max 365 days ahead
cacheSize	Integer	Megabytes (MB)	Integer	0 to 100 MB
floorLevel	Integer	Floor number	Integer	-3 (basement) to 20
phoneNumber	String	E.164 international format	N/A	+[country][number], 8–15 digits

3.3 Performance Requirements

- **[NFR-01]** The system shall load and display the campus map within 3 seconds on Wi-Fi and within 6 seconds on 3G under normal operating conditions.
- **[NFR-02]** The system shall return location search results within 2 seconds for 95% of user queries.
- **[NFR-03]** The system shall compute and display a navigation route within 2 seconds of destination selection for 95% of requests.
- **[NFR-04]** The system shall support a minimum of 500 concurrent active users without degradation in response time.
- **[NFR-05]** The system shall consume no more than 100 MB of device storage for locally cached map and POI data.
- **[NFR-06]** The system shall consume no more than 5% additional device battery per hour during active navigation use.
- **[NFR-07]** The system shall maintain an application crash rate of less than 0.1% of active user sessions.

3.4 Design Constraints

- **[DC-01]** The system shall be developed using Flutter SDK (version 3.x) for a single codebase targeting Android and iOS.
- **[DC-02]** The system shall use Google Maps API or Mapbox as the exclusive mapping service.
- **[DC-03]** The system shall use Firebase Realtime Database as the backend for events and announcement data.
- **[DC-04]** The system shall store all source code in the team GitHub repository using Git version control.
- **[DC-05]** The system shall not store any PII on remote servers; location data used in-session only.
- **[DC-06]** The system shall comply with WCAG 2.1 Level AA accessibility standards for all UI components.

3.5 Software System Attributes

3.5.1 Reliability

The CampusGuide application shall maintain 99% functional uptime during the academic semester. Core features — map display, search, and offline mode — shall remain operational even when remote services are temporarily unavailable by falling back to cached data. The application shall not crash in response to invalid inputs or temporary network loss.

3.5.2 Availability

The application shall be available 24/7 throughout the academic year. Planned maintenance of backend services shall not exceed 2 hours per calendar month, and users shall be notified via an in-app banner at least 24 hours in advance.

3.5.3 Security

- **[SEC-01]** The system shall transmit all data exclusively over HTTPS / TLS 1.2 or higher.
- **[SEC-02]** The system shall not collect, log, transmit, or store any PII including user names or GPS history.
- **[SEC-03]** The system shall validate and sanitise all user-supplied text inputs to prevent injection attacks.
- **[SEC-04]** The system shall store no sensitive API keys in the source code repository; all keys managed via environment configuration files excluded from version control.

3.5.4 Maintainability

The application shall follow clean architecture principles with clear separation across Presentation, Domain, and Data layers. Each module shall be independently unit-testable with a minimum 70% test coverage. All functions and classes shall include inline documentation following Dart documentation conventions.

3.5.5 Portability

A minimum of 90% of the codebase shall be platform-independent Flutter/Dart code, with OS-specific code limited to permission handling. The application shall be deployable to both the Google Play Store and Apple App Store without separate development branches.

3.6 Other Requirements

- **[OTH-01]** The system shall display all content in English as the default and only language in Version 1.0.
- **[OTH-02]** The system shall provide an About screen stating the application version, developing institution, and a link to the GitHub repository.
- **[OTH-03]** The system shall include a first-launch tutorial overlay guiding new users through the four main features: Map, Search, Events, and Emergency Contacts.

Appendix A: Campus POI Data Structure

The following table defines the data fields maintained for each Point of Interest (POI) in the campus database.

Field Name	Data Type	Format / Range	Required	Description
POI ID	String	UUID	Yes	Unique system-generated identifier for each POI
Name	String	Max 100 chars	Yes	Full display name of the location
Category	Enum	See category list	Yes	Type of location (Lecture Hall, Lab, Office, etc.)
Building/Block	String	Max 50 chars	Yes	Name or code of the building or block
Floor	Integer	0 (Ground) – 10	Yes	Floor level within the building
Room Number	String	Alphanumeric, 10 chars	No	Room identifier (e.g., 3B-204)
Department	String	Max 100 chars	Yes	Owning department or faculty
GPS Latitude	Decimal	-90.0 to 90.0	Yes	WGS84 latitude coordinate of POI entrance
GPS Longitude	Decimal	-180.0 to 180.0	Yes	WGS84 longitude coordinate of POI entrance
Description	String	Max 500 chars	No	Brief description of the location and its purpose
Facilities	List	String array	No	Available facilities (e.g., Projector, Whiteboard, Wi-Fi)
Image URL	String	Valid URL	No	Link to a photo of the building or room entrance
Active Status	Boolean	true / false	Yes	Whether the POI is visible to app users

Field Name	Data Type	Format / Range	Required	Description
QR Code	String	Base64 encoded PNG	Auto	System-generated QR code for physical posting

Appendix B: Requirements Traceability Matrix

This matrix lists all functional requirements for traceability and verification purposes.

Req. ID	Requirement Summary	Feature	Priority
FR-MAP-01	Display interactive vector campus map with all buildings and paths	Campus Map	Essential
FR-MAP-02	Support pinch-to-zoom and pan gestures on the map	Campus Map	Essential
FR-MAP-03	Display real-time user location updating every 5 seconds	Campus Map	Essential
FR-MAP-04	Display tappable POI icons for all registered locations	Campus Map	Essential
FR-MAP-05	Display a map legend for POI categories	Campus Map	Essential
FR-MAP-06	Allow users to toggle map layers on or off	Campus Map	Conditional
FR-MAP-07	Provide a 'Recenter' button to return map to user's location	Campus Map	Essential
FR-SCH-01	Text search querying POI name, department, room code, and category	Search	Essential
FR-SCH-02	Display live search results within 500ms of each keystroke	Search	Essential
FR-SCH-03	Rank results by relevance, prioritizing exact name matches	Search	Essential
FR-SCH-04	Support fuzzy matching to handle common misspellings	Search	Conditional
FR-SCH-05	Show 5 most recently searched locations before typing	Search	Conditional
FR-SCH-06	Display 'No results found' for empty search results	Search	Essential
FR-NAV-01	Calculate and display shortest walking route to any POI	Navigation	Essential
FR-NAV-02	Display route as highlighted overlay on campus map	Navigation	Essential

Req. ID	Requirement Summary	Feature	Priority
FR-NAV-03	Display step-by-step written directions in a scrollable panel	Navigation	Essential
FR-NAV-04	Show estimated walking distance and time for route	Navigation	Essential
FR-NAV-05	Auto-recalculate route on deviation within 5 seconds	Navigation	Essential
FR-NAV-06	Display arrival notification within 15m of destination	Navigation	Essential
FR-NAV-07	Provide accessible 'Cancel Navigation' button during navigation	Navigation	Essential
FR-DET-01	Display full POI details: name, category, building, floor, room, dept	Location Detail	Essential
FR-DET-02	Display thumbnail image of building/room entrance for each POI	Location Detail	Conditional
FR-DET-03	Display department contact info on location details screen	Location Detail	Conditional
FR-DET-04	Include 'Get Directions' button on details screen	Location Detail	Essential
FR-DET-05	Include 'Save to Favourites' button on details screen	Location Detail	Essential
FR-QR-01	QR code scanner accessible from main map screen	QR Code	Essential
FR-QR-02	Decode QR code and display POI details within 3 seconds	QR Code	Essential
FR-QR-03	Each POI has a unique, downloadable QR code for printing	QR Code	Essential
FR-QR-04	Display error message for unrecognized QR codes	QR Code	Essential
FR-FAV-01	Allow users to save any POI as a favourite	Favourites	Essential
FR-FAV-02	Automatically record 10 most recently viewed POIs	Favourites	Conditional
FR-FAV-03	Tapping favourite/recent opens POI details and centers map	Favourites	Essential
FR-FAV-04	Allow users to remove items from Favourites list	Favourites	Essential
FR-FAV-05	Persist Favourites and Recents between app sessions	Favourites	Essential
FR-DIR-01	Browsable directory of all POIs organized by category	Directory	Essential

Req. ID	Requirement Summary	Feature	Priority
FR-DIR-02	Each directory category shows count of locations	Directory	Conditional
FR-DIR-03	Tapping directory location opens details and centers map	Directory	Essential
FR-DIR-04	Directory searchable via main search bar	Directory	Essential
FR-DIR-05	Directory available offline using cached data	Directory	Essential
FR-ADM-01	Web-based admin panel with authenticated access	Admin Panel	Essential
FR-ADM-02	Admin can create new POI records with all required fields	Admin Panel	Essential
FR-ADM-03	Admin can edit any field of an existing POI record	Admin Panel	Essential
FR-ADM-04	Admin can deactivate a POI without deleting it	Admin Panel	Essential
FR-ADM-05	Admin can generate and download printable QR code for each POI	Admin Panel	Essential
FR-ADM-06	All admin changes are logged with user identity and timestamp	Admin Panel	Essential